Shanay Ambani

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Education

Purdue University | *Bachelor of Science in Mechatronics Engineering Technology* 05/2026

- **Coursework:** Mechanics & Materials, Manufacturing Processes, Automation & Control, CAD & Design, LabVIEW, Instrumentation & Data Acquisition, Digital Systems, Physical Foundations, Leadership & Professional Development.

Skills

- **Technical:** SolidWorks, CAD (AutoCAD, Inventor), GD&T, Engineering Drawings, Mechanical Design, Design for Manufacturability (DFM), Manufacturing Processes, Rapid Prototyping, Prototype Fabrication, Structural Analysis (FEA), Tolerance Analysis, 3D Scanning (MeshMixer, QuickSurface), LabVIEW, Microsoft Excel, Microsoft PowerPoint.

Experience

D&O Motorsports | *Design Engineering Intern* 05/2025 - 08/2025

- Reduced drag by 4% by redesigning the Honda City bumper using SolidWorks and validating results with CFD analysis.
- Developed manufacturable CAD models from scan data using 3D scanning and surface reconstruction techniques.
- Designed and prototyped aerodynamic components within 0.5 mm tolerance using CAD and fabrication methods.
- Improved manufacturability and performance through iterative testing and design refinement.

L.I.M.E (Launchpad for Indian Motorsport Engineers) | *Fellow and Intern* 05/2024 - 08/2024

- Completed an 80 hour motorsport engineering program focused on CAD modeling, prototyping, and mechanical systems.
- Selected as one of 70 fellows based on strong technical performance and engineering problem solving ability.
- Improved CAD accuracy by 10% using 3D scanning and surface refinement for prototype development.
- Modified component geometry to support manufacturability and system integration.

Projects

Senior Design Capstone (Purdue University) | *Mechanical Design Lead* 08/2025 - Present

- Designed a mounting system for an RFID platform achieving 21 ft detection coverage from a 7 ft height.
- Created CAD assemblies and engineering drawings to support fabrication, testing, and system integration.
- Performed structural analysis under 8.7 lb ft torque and 30 mph wind conditions to validate design reliability.
- Improved system reliability by 18 percent through alignment, stiffness optimization, and iterative prototyping.

Skoda Superb Rear Spoiler Concept | *Shadow Intern* 12/2024 - 01/2025

- Assisted in designing a rear diffuser and spoiler system to improve aerodynamic stability and manufacturability.
- Modeled components and evaluated airflow behavior using CAD modeling and CFD tools.
- Applied insights to support development of a front aero concept for prototype evaluation.

Formula SAE Selection Project (Chain Drive System – Purdue University) 09/2024 - 10/2024

- Modeled a chain drive system including sprocket geometry, tensioning, and load transfer using CAD.
- Evaluated alignment and clearance requirements to ensure manufacturability and reliable operation.
- Produced engineering drawings and documentation to support fabrication and design validation.

Academic Papers

Water Hardness Study (Chemistry – International Baccalaureate – Individual Assessment)

- Identified an estimated 7% reduction in water hardness by analysing titration results across 5 temperatures.

Engine Oil Viscosity Study (Physics – International Baccalaureate – Individual Assessment)

- Measured engine oil viscosity under varying temperatures and observed an estimated 20% decrease as temperature increased.

Volunteering

Step Up NGO | *Co-Founder* 05/2020 - Present

- Co-founded a nonprofit supporting holistic development for underprivileged children.
- Secured over $11000 through relief drives and crowdfunding, supporting more than 200 students.
- Expanded outreach to over 200 children through sports and environmental initiatives.

Interests

Swimming, Cricket, Photography (featured on Animal Planet India), Top 10% Wharton Global Youth Investment Competition.